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25-Item AI Literacy Test (AILT)

Question

Note. Order of questions and order of answers within each question are randomized.

1. Which of the languages below are used to code a robot?

- a) Python [correct]
- b) HTML
- c) Computer Vision
- d) Hypertext

2. Which technology is the primary enabler of Artificial Intelligence?

- a) Machine Learning [correct]
- b) Electric Battery
- c) Robotics
- d) Engineering

3. Which of the following tasks was the first to be performed well by Artificial Intelligence?

- a) The complete replacement of a doctor in the treatment of patients
- b) The implementation of psychotherapies
- c) The analysis of X-ray images, for example to detect a torn meniscus or a tumor [correct]
- d) The performance of complex physical work that requires dexterity or precise hand-eye coordination

4. Which of these is a notable algorithmic problem?

- a) Counting
- b) Sorting [correct]
- c) Multiplication
- d) Division

5. What was the purpose of the PCI DSS (payment card industry data security standard)?

- a) Regulate financial transactions [correct]
- b) Manage browser cookies
- c) Standardize credit score data
- d) Enable websites to accept crypto currencies

6. Which of these is not a best practice when trying to protect your privacy online?

- a) Using a two-factor authentication to access account and devices
- b) Storing your passwords in the notes section of your smartphone [correct]
- c) Updating your operating system promptly when an update becomes available
- d) Using different login information

7. If using an algorithm, can the results be biased against specific groups of people?

- a) No, unlike humans, algorithms are free of emotions and cannot be wrong
- b) Yes, but only if you train them on privileged populations
- c) Yes, but only if you use them with incomplete data
- d) Yes, even with correct data and training, algorithms can provide biased results [correct]

8. How algorithmic decisions best explained?

- a) Only for very basic and simple algorithms
- b) They can be explained by sharing the algorithm with everyone
- c) As complex code, they cannot be explained
- d) Algorithms can be explained by sharing the contributing factors to decisions [correct]

9. When programming stock investment recommendation algorithms, these algorithms are typically programmed to:

- a) Ignore smaller companies
- b) Give more weight to random fluctuations in the market
- c) Consistently keep track of investments [correct]
- d) Prefer indices over specific stocks

10. According to the Health Insurance Portability and Accountability Act of 1996 (HIPAA), protected health information cannot be shared with:

- a) The patient's employer [correct]
- b) The patient
- c) Law enforcement
- d) The patient's health insurance

11. AI can take images and short sound clips of people and create long, convincing impersonation of these people saying anything. What is this technology called?

- a) Deepmind
- b) Deepfake [correct]
- c) Deeplearn
- d) Deepvoice

12. Google automatically completes the text you type in the search bar...

- a) By looking at the most common search terms used by others and your search history [correct]
- b) Based only on your search history
- c) Based only on what people in your zip code are searching for
- d) By using your phone camera when you type and reading your lips

13. Algorithms may improve website performance by doing all of the following except for....

- a) Showing us content that might interest us
- b) Showing us relevant ads
- c) Monitoring which content we engage in
- d) Changing our screen brightness [correct]

14. What information do social media algorithms not typically use when deciding what information to display to a person?

- a) Articles people have previously clicked on
- b) Videos people have previously watched online
- c) Information people have shared on social media
- d) What would benefit society [correct]

15. When you use a web-based service which stores its data in "the cloud", where is the data actually stored?

- a) In a hidden folder in your computer
- b) At the company's headquarters
- c) At a large server farm [correct]
- d) In your browser's cookies

16. If you turn off personalized ads, you are most likely to see ads...

- a) Targeting people similar to you
- b) That are relevant for you
- c) That are interesting to you
- d) For commonly purchased products [correct]

17. Pattern recognition is a decision-making step employed by...

- a) Algorithms
- b) Humans
- c) Both [correct]
- d) Neither

18. When do algorithms tend to perform better than humans?

- a) When lots of structured data is involved and the risk from a wrong answer is minimal [correct]
- b) When there is no structured data available and the risk from a wrong answer is minimal
- c) When lots of structured data is involved and the risk from a wrong answer is significant
- d) When there is no structured data available and risk from a wrong answer is significant

19. What is training data and why is it important?

- a) Training data is used when the model is fully calibrated, to train new users with how to work with the model
- b) Training data is used to calibrate the model during creation, creating a model that performs best according to that data [correct]
- c) Training data is used to test the model after it was created, to see how it works with new data
- d) Training data is used to minimize bias in older models and train them to new population types

20. Indicate if the following is primarily the result of an algorithm or not:

What I write in a post on social media

- a) Yes
- b) No [correct]

21. Indicate if the following is primarily the result of an algorithm or not:

What I type in a text message to a friend

- a) Yes
- b) No [correct]

22. Indicate if the following is primarily the result of an algorithm or not:

When I read a direct message on a social media platform

- a) Yes
- b) No [correct]

23. Indicate if the following is primarily the result of an algorithm or not:

Returning the results of a search query on a search engine (e.g., Google)

- a) Yes [correct]
- b) No

24. Indicate if the following is primarily the result of an algorithm or not:

Dual authentication to access one of my accounts

- a) Yes [correct]
- b) No

25. What is the process of a machine learning model making a prediction based on new data called?

- a) Coding
- b) Training
- c) Testing
- d) Inference [correct]